

Do Classical Price-Action Events Carry Directional Information?

Test A of a Role-Structured Confluence Study

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Abstract

Traders often claim that overlapping technical patterns—“confluence”—produce stronger price reactions than isolated signals. We pre-registered and executed Test A of a staged screening study: 76 price-action event types across four timeframes (284 cells) on EURUSD (2015–2021 screen split). A critical methodological finding emerged: uniform random-time controls were invalid on M15 due to session-volatility confounding (random forward movement varied $3.7\times$ by hour-of-day). After amendment to hour-matched controls, five cells survived screening (all M15). One confirmed on frozen 2022–2024 data (`trendline_liquidity_sweep_low`, +0.30 ATR effect); one replicated on frozen GBPUSD (`channel_top_touch`). Effects are real but small—suitable as gate inputs, not standalone strategies. This repository is observation-only and does not trade.

1 Introduction

1.1 Research question

H_0 : Conditional on a classical price-action event, ATR-normalised directional forward excursion is indistinguishable from direction- and time-matched random baselines.

H_1 : Specific (timeframe $\times$ event type) cells beat that baseline; specific cross-timeframe combinations beat their parts.

This is Test A (price action only) of a three-family program (Tests B: indicators; C: cross-family interactions). Test A tests each method individually by timeframe before any combination claims.

2 Methodology

2.1 Event dictionary (Stage 0)

Eighteen detector modules emit 76 causal event types: market structure (BOS/CHoCH), zones/order blocks/breakers, fair-value gaps, liquidity pools and sweeps, trendlines/channels, horizontal S/R, chart patterns, fibonacci tags, and candlestick patterns. Each event carries a pre-registered directional hypothesis (+1 up / -1 down). The dictionary was frozen before any statistics were computed.

2.2 Outcome metric

For an event at bar t : maximum favourable excursion (MFE) in the event direction over horizon H (D1: 30, H4/H1: 20, M15: 16 bars), divided by ATR(14); plus binary hit = reaches +1 ATR before -1 ATR.

2.3 Controls and amendment v2.1

Pre-registered controls: random bar indices with directions resampled from the event mix. This produced 41/284 “significant” cells—including mutually contradictory hypotheses—indicating a broken null. Diagnostic analysis showed MFE at random M15 times varies from 1.09 ATR (19:00 UTC) to 4.03 ATR (05:00 UTC) because ATR(14) lags the intraday session cycle.

Amendment v2.1 (documented before re-run): controls matched to each event’s hour-of-day and direction. The invalid uniform run is retained as a cautionary record.

2.4 Statistics

Permutation test (2,000 shuffles), one-sided; Benjamini–Hochberg FDR at 5% across adequately powered cells ($n \geq 100$). Four-tier verdicts: **alive**, **parked_weak_effect**, **parked_insufficient_n**, **dead**.

2.5 Data splits

Split	Range	Use
Screen	2015-01-01 – 2021-12-31	selection
Confirm	2022-01-01 – 2024-12-31	frozen test of survivors
Sealed	2025-01-01+	untouched
Cross-pair	GBPUSD	frozen replication

3 Results

3.1 Stage 1 (hour-matched controls)

284 cells screened: **5 alive**, 16 parked weak, 77 insufficient- n , 186 dead. All survivors on M15:

Event type	n	Effect (ATR)	Screen p
channel_top_touch	19,793	+0.075	0.0005
fib_50_tag	5,134	+0.152	0.0005
trendline_liquidity_sweep_low	1,133	+0.303	0.0005
fib_618_tag	4,676	+0.124	0.0010
fib_ext_1272_tag	2,482	+0.189	0.0010

3.2 Out-of-sample confirmation

- **Confirm split:** trendline_liquidity_sweep_low confirmed (+0.308 ATR, $p = 0.0065$).
- **GBPUSD replication:** channel_top_touch replicated (+0.099 ATR, $p = 0.0005$).
- All five survivors positive on both OOS tests (sign-consistency $\approx 3\%$ under random-sign null).

3.3 Stage 2

Strict Stage 2 (alive cells only) was **empty by construction**—no higher-timeframe survivor to serve as context. Exploratory Stage 2 (including parked-weak cells) found most “context $\times$ setup” lift explained by setup marginals; H1 **equal_highs_pool** context flagged for pre-registered Stage-2b.

4 Discussion

The session-volatility confound is a general lesson for intraday FX research: ATR-normalised metrics on M15 require time-matched nulls. Surviving effects (+0.05 to +0.35 ATR MFE) are too small for standalone M15 strategies after spread but may inform confluence gating or exit logic in a validated main agent—only via that agent’s separate pipeline.

5 Limitations

- Detector definitions are one auditable formalisation per concept.
- D1 cells are structurally underpowered at $n \geq 100$ per pair.
- MFE measures opportunity, not net P&L; costs not modelled.
- Tests B and C not yet pre-registered.

6 Conclusion

Classical price-action confluence is **partially supported** on M15 trendline/channel geometry, not broadly across the full dictionary. The honest headline: a handful of modest, reproducible signals—not the “everything works on M15” illusion the broken first analysis suggested. This lab never trades; findings are hypothesis-generating for external validation.

Artifacts

Registries and summary figure: `output/`. Full narrative: `REPORT.md`. Protocol: `PROTOCOL.md`.